4.8 Analyzing Graphs of Polynomial Functions

Name _____ Period ____ Date _____

Algebra II 4.8 · Student sheet

I can…

- use x-intercepts and end behavior to graph polynomial functions.
- find turning points and local maximums and minimums.
- tell whether a function is even, odd, or neither.

Vocabulary · fill in the missing word

Local maximum / minimum — the _____ of a turning point that is higher / lower than all nearby points

Turning points — a _____ of degree n has at most n − 1 turning points

Even function — f(−x) = f(x); _____ about the y-axis

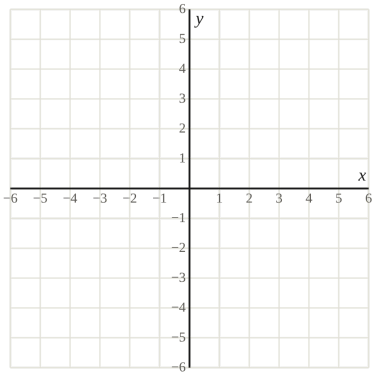
Odd function — f(−x) = −f(x); _____ about the origin

Multiplicity — the number of times a factor _____; even multiplicity → the graph touches the x-axis, odd → it crosses

Part 1 · Sketch from factors

y = (x + 2)(x − 1)(x − 3). x-intercepts: _____ End behavior: _____

Sketch first (no technology):



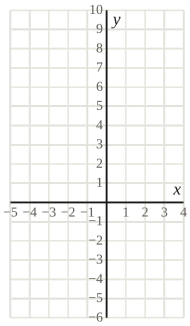
Now check in Desmos. Local maximum ≈ _____ Local minimum ≈ _____

Part 2 · Notes and examples (fill in each note, then work the example)

Multiplicity: Even multiplicity: the graph _____ the x-axis. Odd: it _____.

Example 1

Graph $f(x) = \frac{1}{2}(x + 3)(x - 2)^2$.



Example 2

Find all real zeros of $f(x) = x^3 - x^2 - 6x$.

Turning points: A degree-n polynomial has at most _____ turning points.

Example 3

A polynomial graph has 3 turning points. What is its least possible degree?

Even and odd functions: Even function: symmetric about the _____. Odd function: symmetric about the _____.

Example 4

Tell whether each function is even, odd, or neither.

a. $f(x) = x^4 - 3x^2$ b. $g(x) = 2x^3 + x$ c. $h(x) = x^3 + 2$

Part 3 · Partner practice

Partner 1

Find all real zeros of $f(x) = x^4 - 9x^2$.

Partner 2

Is $f(x) = x^5 - 4x^3$ even, odd, or neither?